

A shopping mall plans to develop a Smart Vehicle Parking System to automate vehicle entry, slot allocation, billing, and exit control. The system will be used by drivers, parking attendants, and an external payment gateway. The mall management wants the development team to first perform proper Requirements Engineering before proceeding to design.

The system should support Vehicle entry and registration, Automatic slot allocation, Real-time slot availability display, Parking fee calculation based on duration, Digital payment at exit, Gate control after successful payment, Administrative monitoring and reporting.

- a) As part of Requirements Engineering, identify and describe the major system events, corresponding system operations, and the key conceptual classes involved in the above scenario. (4 marks)
- b) Explain, with reference to the above scenario, how functional modeling using classes and their responsibilities helps in identifying system operations and constructing the System Sequence Diagram. (5 marks)
- c) Using the identified events, operations and classes, draw a System Sequence Diagram (SSD) for the use case “Vehicle Entry and Slot Allocation”. (5 marks)
- d) Analyze the above scenario and explain how unclear or incomplete requirements (e.g., handling peak-hour traffic, payment failures, or slot overbooking) could impact the identification of system events, operations and classes during the Requirements Engineering phase. (2 marks)